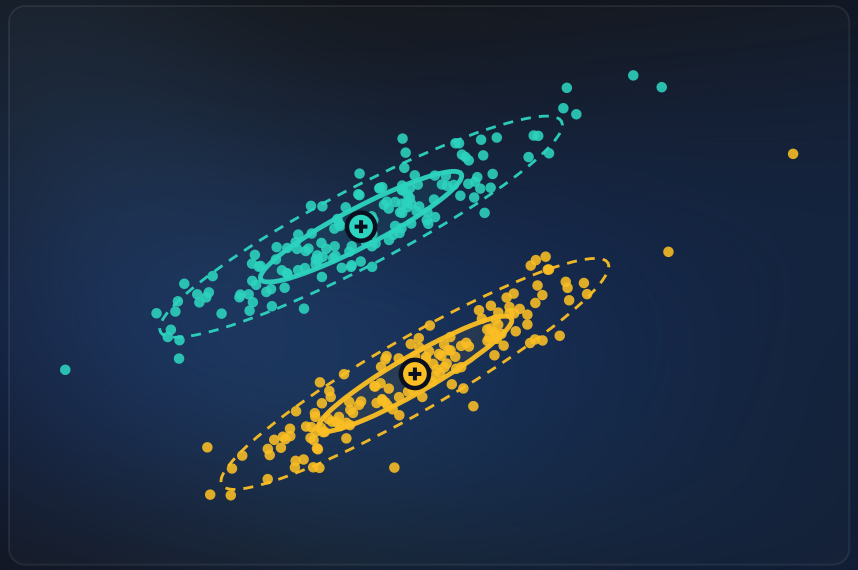
Gaussian Mixture Models

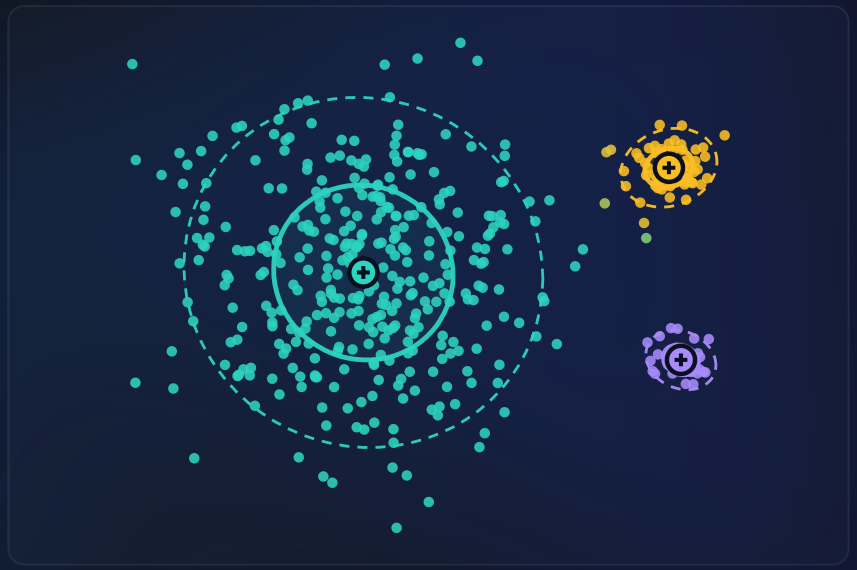
Clusters as overlapping, probabilistic ellipses

Unsupervised Learning · Foundations of Machine Learning

Where k-Means Left Off



Stretched clusters · GMM ARI 1.00



Unequal sizes · GMM ARI 0.97

Replace each **center** with a **Gaussian** (its own mean, spread and tilt), and each **hard label** with a **probability** of membership.

Today

1 · One Gaussian = one cluster

Mean, covariance, and the ellipse they draw.

2 · Mixtures and responsibilities

How the data was generated, and Bayes' rule for who generated each point.

3 · Fitting with EM

Alternate soft labels and refits. Why it climbs, where it gets stuck.

4 · Model choices

Covariance type and number of components, with BIC.

5 · Real data, and what a density buys you

Old Faithful and iris · anomaly detection · generating new data.

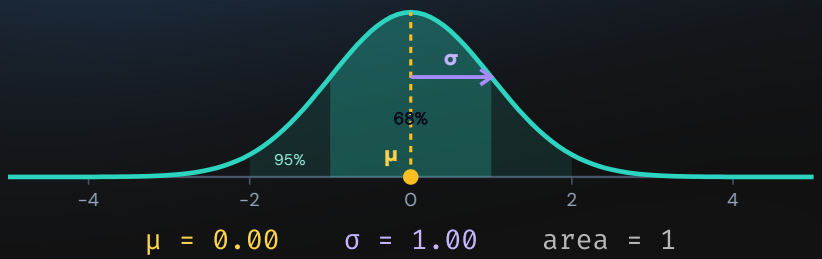
6 · The big picture

k-means as a special case, and when to use which method.

Part 1 · One Cluster, One Gaussian

A center is a point. A Gaussian is a center plus a shape.

The 1D Gaussian

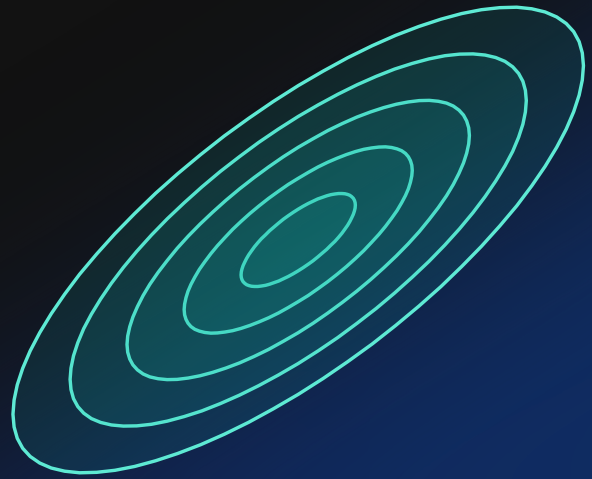


$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\,\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

μ moves the bell. σ widens it, and the peak drops so the area stays 1.

68% of the mass within 1σ, 95% within 2σ. Density falls off with the **squared, standardized** distance from μ.

The 2D Gaussian Is a Hill



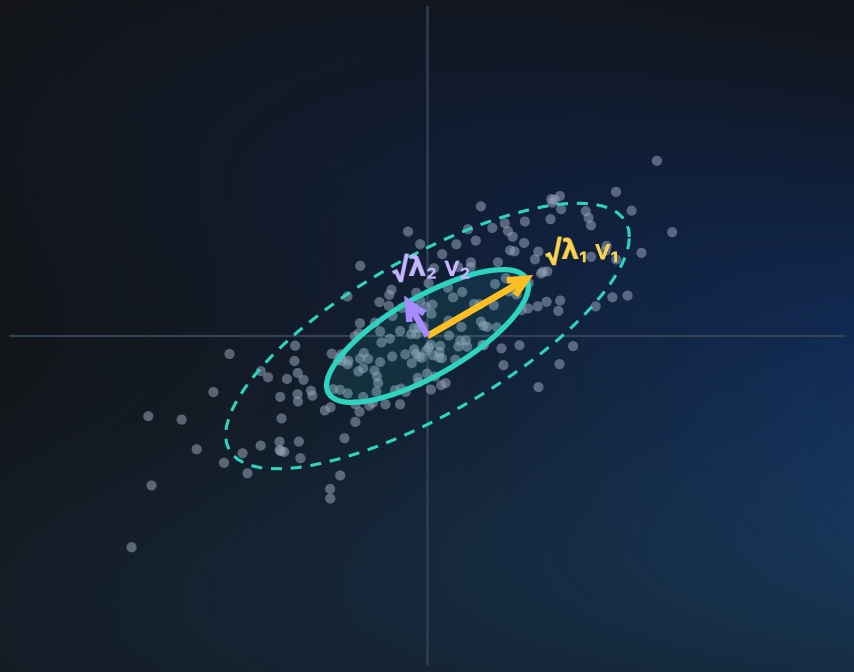
Spherical $\Sigma = I$ · circles

Diagonal $\Sigma = \mathrm{diag}(2.4,\ 0.45)$ · axis-aligned

Full $\Sigma = \begin{bmatrix} 1.8 & 1.1 \\ 1.1 & 1.2 \end{bmatrix}$ · tilted

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}\,|\Sigma|^{1/2}}$$

Covariance Is Geometry



Σ (LIVE)

$$\begin{bmatrix} 2.34 & 1.15 \\ 1.15 & 1.01 \end{bmatrix}$$

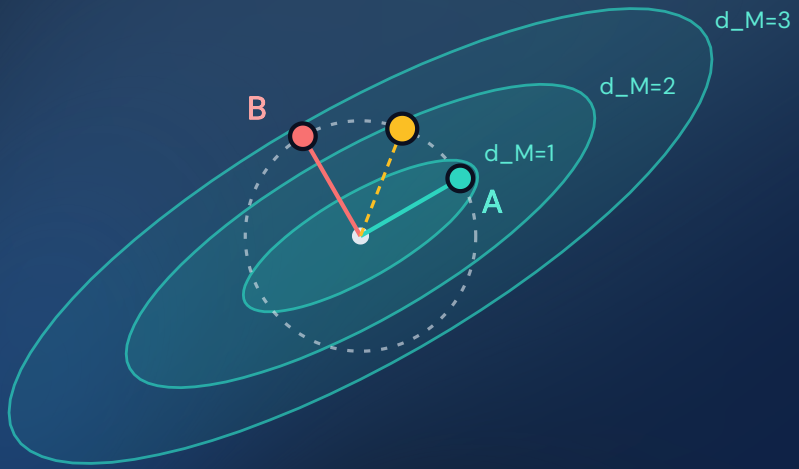
$\lambda_1 = 3.00 \cdot \lambda_2 = 0.35$

tilt = 30° · corr ρ = 0.75

Diagonal entries: spread along each axis. Off-diagonal: tilt (correlation). Eigenvectors: the ellipse's axes; √eigenvalues: its radii.

$$\Sigma = V\Lambda V^\top = \lambda_1 \mathbf{v}_1\mathbf{v}_1^\top + \lambda_2 \mathbf{v}_2\mathbf{v}_2^\top \qquad \text{axes } \mathbf{v}_j, \ \text{radii } \sqrt{\lambda_j}$$

Distance the Gaussian Way: Mahalanobis



Euclidean	1.50
Mahalanobis	1.84

density / peak

A: d_M = 0.87 · density 69% of peak
B: d_M = 2.74 · density 2.4% of peak

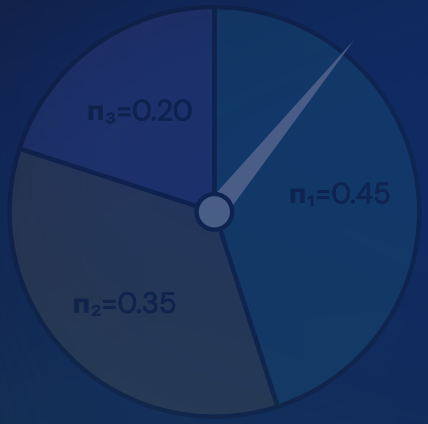
$$d_M(\mathbf{x}) = \sqrt{(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})}, \qquad \mathcal{N} \propto e^{-d_M^2/2}$$

Distance measured in standard deviations **along each direction**. Equal-density curves are ellipses, not circles.

Part 2 · Mixtures and Responsibilities

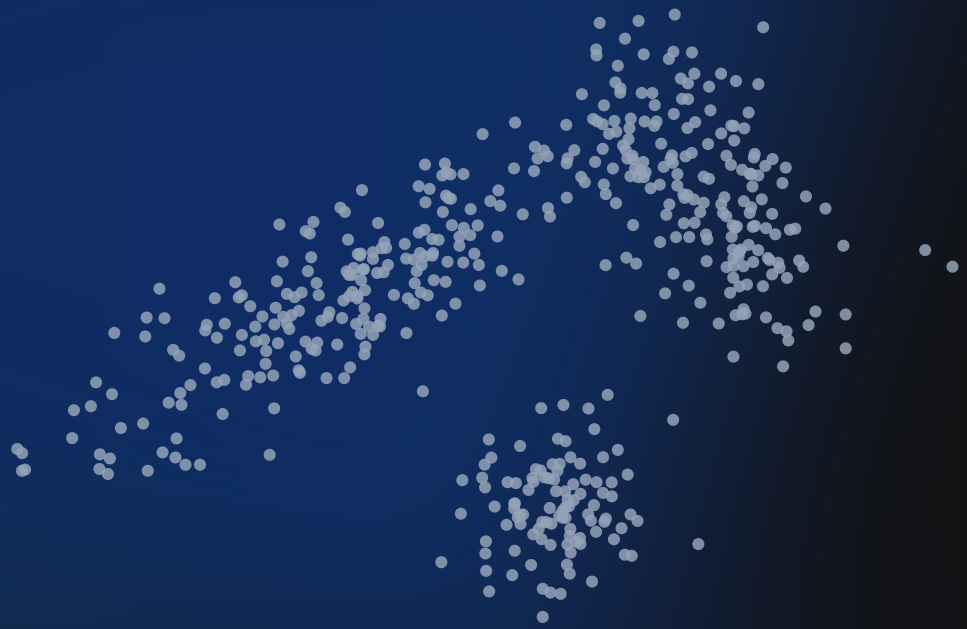
A story for how the data was made, and Bayes' rule to read it backwards

A Mixture Is a Generative Story



500 points drawn

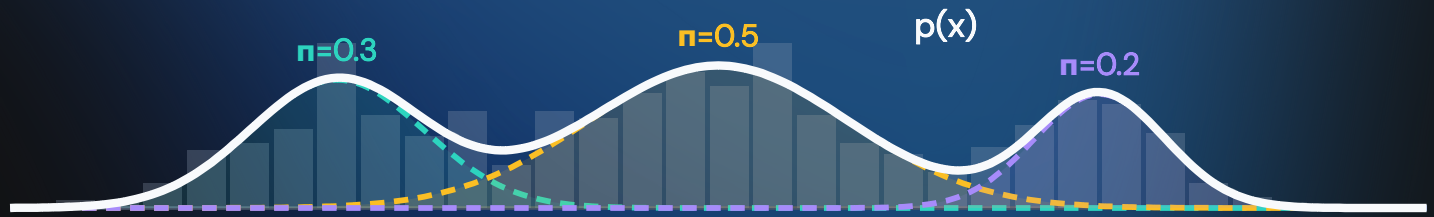
What we observe: no k, no ellipses



$$z_i \sim \text{Categorical}(\pi_1, \ldots, \pi_K), \qquad \mathbf{x}_i \mid z_i = k \;\sim\; \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$$

We only see x. The labels z are hidden (latent): recovering them is the clustering problem.

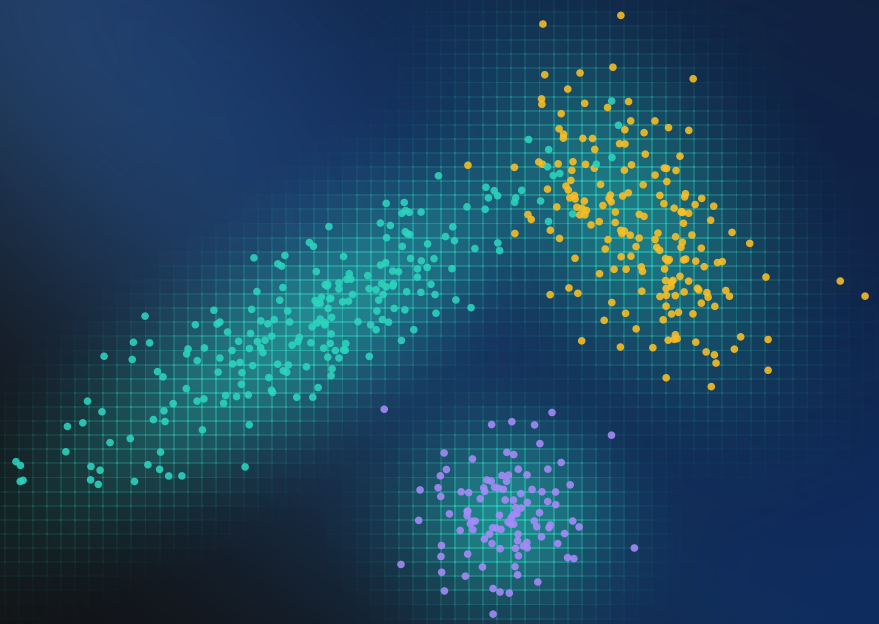
Adding Up Weighted Bells



600 samples from the story land right on it.

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k), \qquad \pi_k \geq 0, \sum_k \pi_k = 1$$

The Mixture as a Landscape



One hill per component. A heavy weight or a small covariance makes a tall hill.

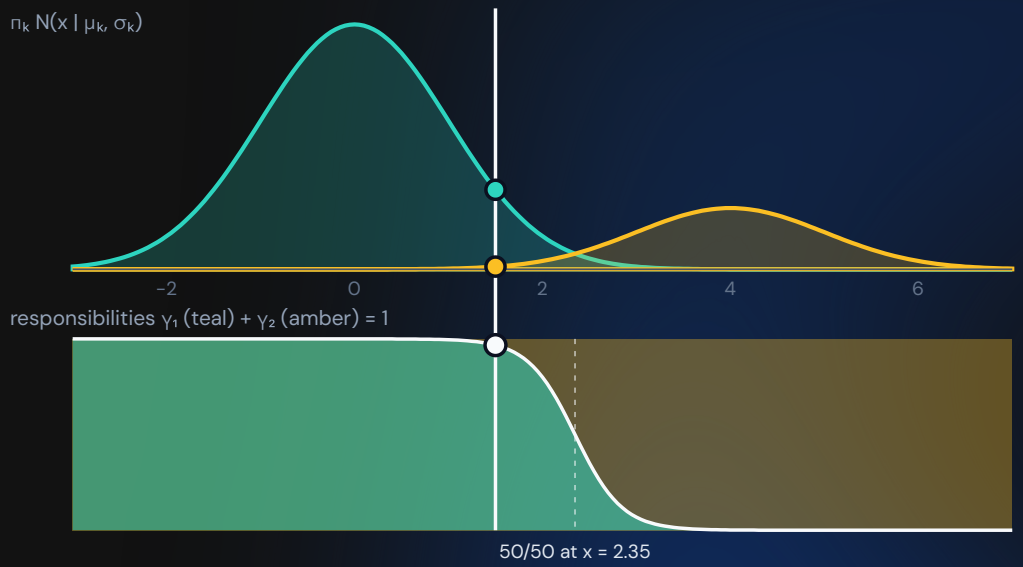
Everything to learn

$\theta = \{ \pi_k, \mu_k, \Sigma_k \}$ for $k = 1 \ldots K$

Here: 2 weights + 6 means + 9 covariance entries = **17 numbers**

A density for **any** point x, not just a label for the training points.

Responsibilities: Bayes' Rule for Membership



x = 1.50 · π = (0.80, 0.20)

$\pi_1 N_1$ = 0.1036
$\pi_2 N_2$ = 0.0035

0.967

γ = each height divided by the sum of the heights

$$\gamma_{ik} = P(z_i = k \mid \mathbf{x}_i) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j)}$$

Raise the prior π_1 to 0.8: same x, now $\gamma_1 \approx 0.97$, and the 50/50 point moves right.

Worked Example: One Responsibility by Hand

Two 1D components: $\pi_1 = \pi_2 = 0.5$, $\mathcal{N}(0, 1)$ and $\mathcal{N}(4, 1)$. Point $x = 1.5$.

- $\mathcal{N}(1.5 \mid 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-1.5^2/2} \approx 0.1295$
- $\mathcal{N}(1.5 \mid 4, 1) = \frac{1}{\sqrt{2\pi}} e^{-2.5^2/2} \approx 0.0175$
- prior × likelihood: $0.5(0.1295) = 0.0648$ and $0.5(0.0175) = 0.0088$
- $\gamma_1 = \dfrac{0.0648}{0.0648 + 0.0088} \approx 0.880, \quad \gamma_2 \approx 0.120$

```python
import numpy as np
from scipy.stats import norm

x = 1.5
pi = np.array([0.5, 0.5])
lik = norm.pdf(x, loc=[0, 4], scale=1)
unnorm = pi * lik      # [0.0648 0.0088]
gamma = unnorm / unnorm.sum()
print(gamma.round(3))  # [0.88 0.12]
```

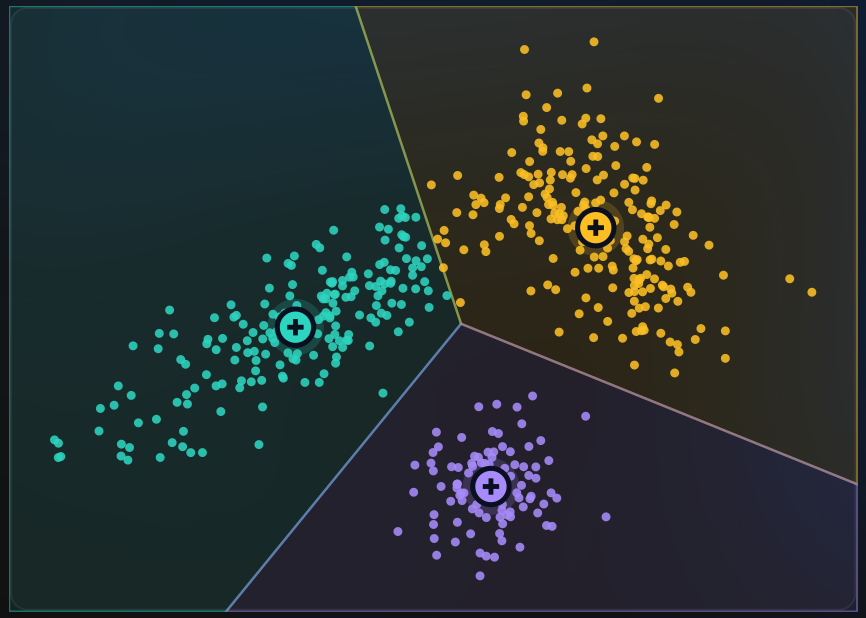
Soft Labels on the Running Example

Soft: each point's color is a blend of the component colors, weighted by its γ.

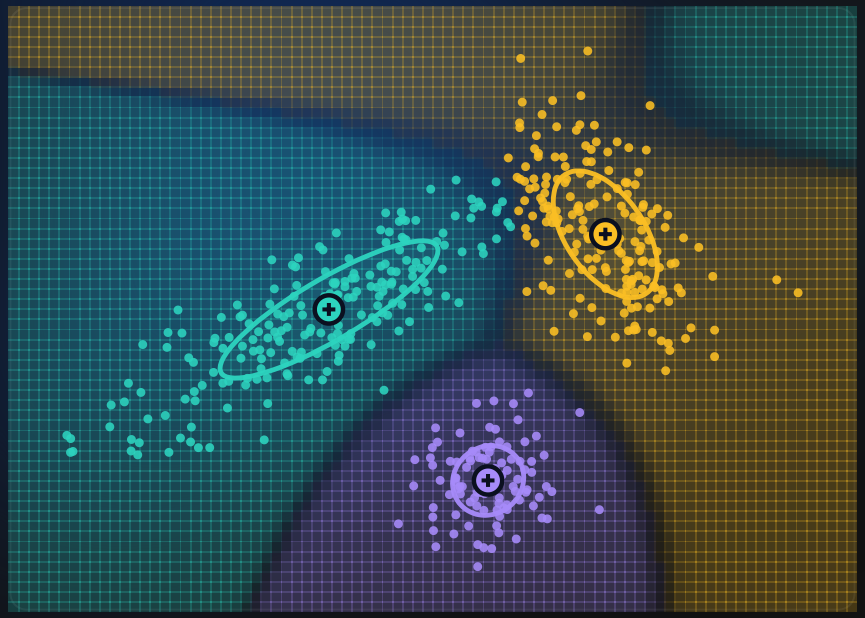
Hard: keep only the largest γ, `predict`. The boundary points snap to one side.

59 of 500 points have no component above 0.9. They sit where ellipses overlap, which is exactly where a hard label would be a guess.

Curved Borders Instead of Straight Ones



k-means · ARI 0.80



GMM · ARI 0.88

k-means: nearest center by Euclidean distance. Borders are perpendicular bisectors, always **straight**.

GMM: largest $\pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$. Comparing two quadratic exponents gives **quadratic** borders: ellipses, parabolas, hyperbolas.

Part 3 · Fitting a Mixture with EM

Guess the labels, fit the Gaussians, repeat

Maximum Likelihood, and Why It's Hard

What we can observe: x only

$$\log L(\theta) = \sum_{i=1}^{n} \log \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)$$

A **log of a sum**: the log can't reach the Gaussians. Setting the gradient to zero gives equations where every μ_k depends on all the others. No closed form.

If we also knew z

$$\log L_c(\theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} z_{ik} \big[\log \pi_k + \log \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k) \big]$$

A **sum of logs**: it splits into K separate problems. Each Gaussian is fit to its own points: sample mean, sample covariance, fraction of points.

EM's trick: we don't know $z_{ik} \in \{0, 1\}$, so replace it with its **expected value** under the current guess, $\mathbb{E}[z_{ik}] = \gamma_{ik}$, then solve the easy problem.

A Chicken-and-Egg Problem

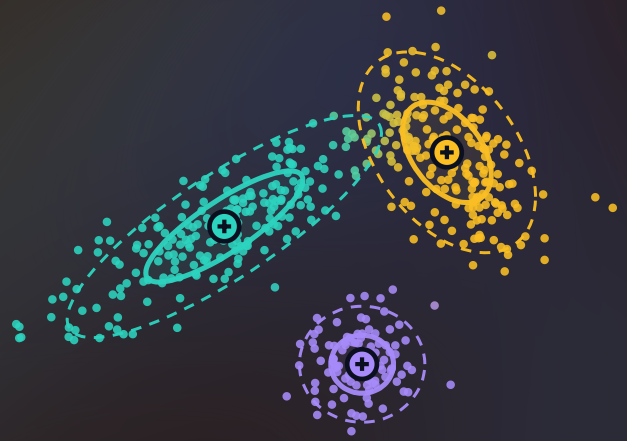
If we knew the labels…

…each Gaussian is just its group's fraction, mean and covariance.

EM

If we knew the Gaussians…

…each point's membership is Bayes' rule: the responsibilities.



Neither is known. Guess one, then alternate: each half-step uses the other's latest answer.

The EM Algorithm

E-step · soft labels

$$\gamma_{ik} \leftarrow \frac{\pi_k \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_j \pi_j \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j)}$$

Parameters fixed. Bayes' rule for every point and component: an n × K table.

M-step · weighted refits

$$N_k = \textstyle\sum_i \gamma_{ik}, \qquad \pi_k = N_k / n$$

$$\boldsymbol{\mu}_k = \tfrac{1}{N_k} \textstyle\sum_i \gamma_{ik} \, \mathbf{x}_i$$

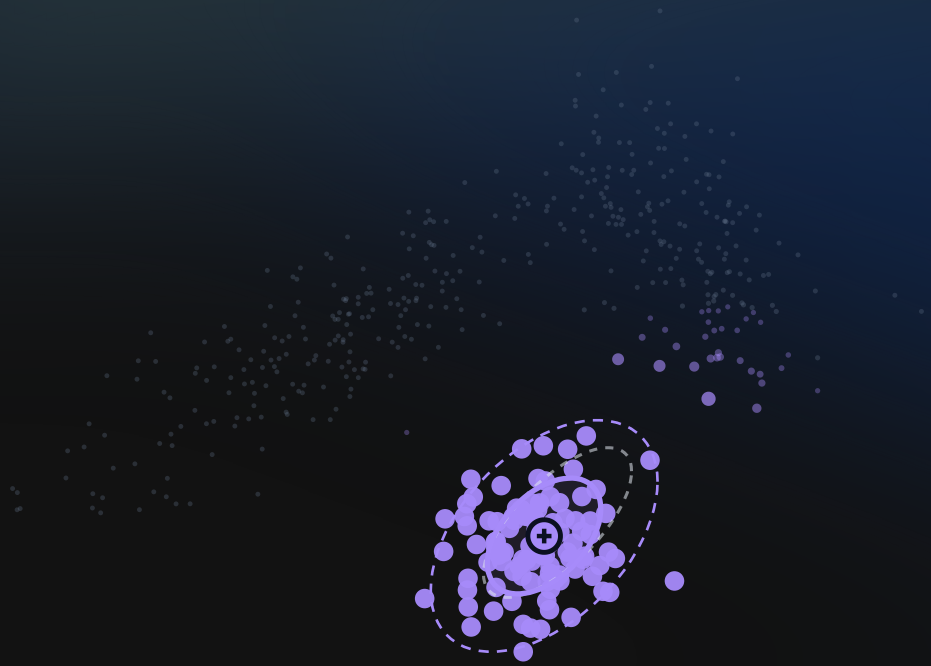
$$\Sigma_k = \tfrac{1}{N_k} \textstyle\sum_i \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top$$

Initialize means (scikit-learn: from k-means), covariances, equal weights.

Repeat E, M until log L improves by less than `tol` per point (1e-3).

Cost per iteration: O(n K d²) for full covariances (plus K matrix inversions).

The M-Step Up Close: Weighted Votes



Component 3

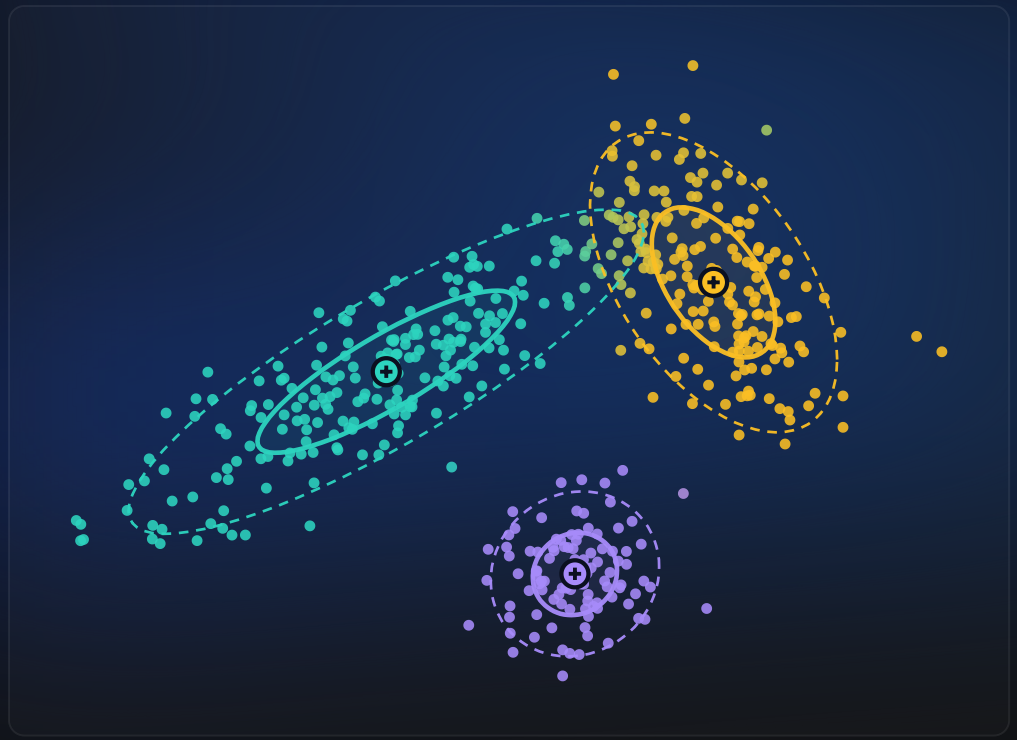
$N_k = \Sigma_i \gamma_{ik} = 108.1$ of 500

$\pi_k = N_k/n = 0.216$

$\mu_k = (2.66, -2.43)$

Big, bright dots pull hard on the new mean and shape; faint dots barely count. Dashed white: the ellipse before this M-step.

Watch EM Run



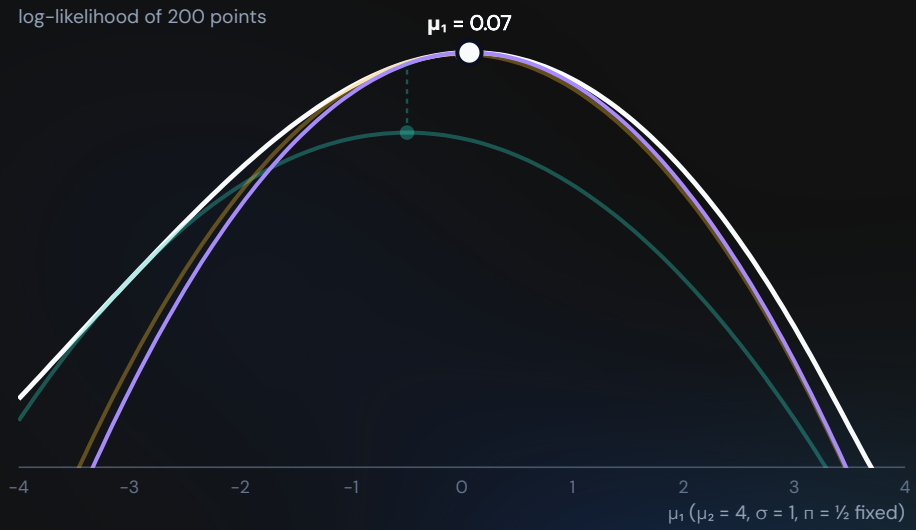
iteration **15**

E-step · recolor points | M-step · refit ellipses

log-likelihood

$\pi_1 = 0.43$ $\pi_2 = 0.36$ $\pi_3 = 0.21$

Why EM Never Goes Downhill

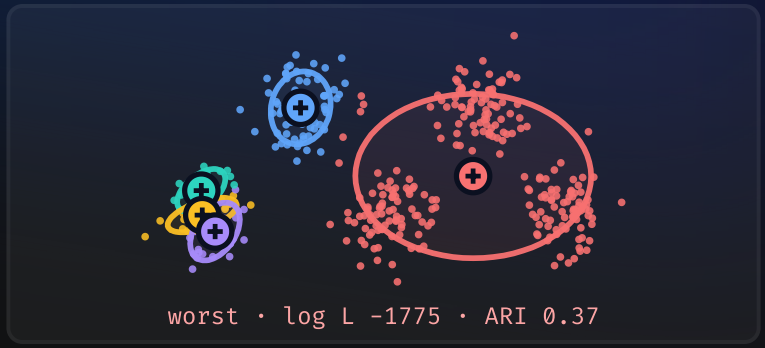


E-step builds a simpler curve, a **lower bound** on log L, that **touches** it at the current θ.

M-step jumps to the top of that bound. log L there is ≥ the bound ≥ log L before.

Third: 0.07. Steps shrink as EM nears the peak.

Local Optima: Restart and Keep the Best



`n_init=10` runs EM from 10 starts and keeps the highest likelihood. Default is 1.

The default k-means start can inherit k-means' mistake: on the stretched clusters, GMM from one k-means start scores ARI **0.03**; 20 random starts, **1.00**.

When the Likelihood Blows Up

σ of component 4 = **0.050**

density at its point = 6.4e+1

log L gained over the 3-component fit

+6.9

`reg_covar` (default 1e-6) adds a floor to every variance, so Σ stays invertible and the spike stays finite. Floor exaggerated here.

Part 4 · Model Choices

How flexible is each Gaussian, and how many are there?

Four Covariance Types

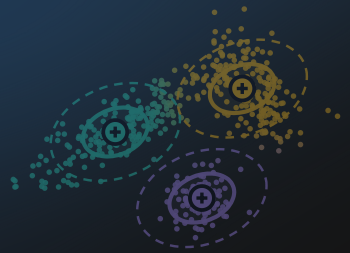
"spherical"
one variance per component: circles

p	BIC	ARI
11	3745	0.81

"diag"
one variance per axis: axis-aligned ellipses

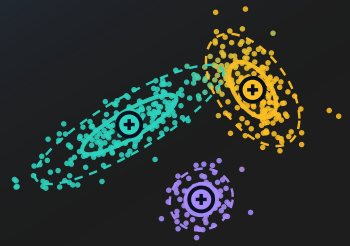
p	BIC	ARI
14	3729	0.81

"tied"
one shared full Σ: same shape everywhere



p	BIC	ARI
11	3811	0.79

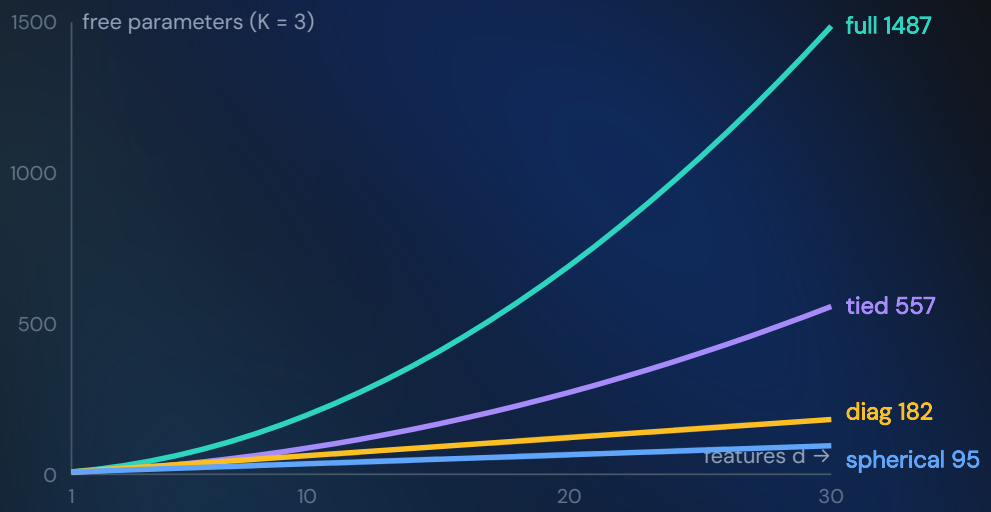
"full"
own full Σ each: any ellipse



p	BIC	ARI
17	**3485**	0.88

full: each component its own ellipse. Best BIC here, and the default.

Flexibility Costs Parameters

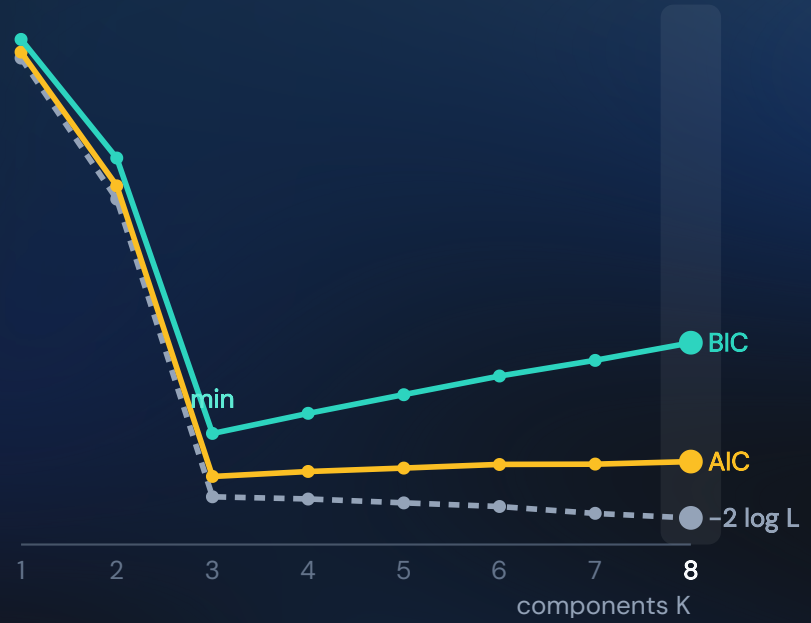


full grows like K·d²/2: each component needs d(d+1)/2 covariances. 64 features, 3 components: **6,434** parameters.

Few points per component + many parameters → noisy or **singular** covariances, overfitting.

diag / **spherical** act as regularization: less flexible, far fewer numbers to estimate. Or reduce d first (next deck).

Choosing K: Penalize the Parameters

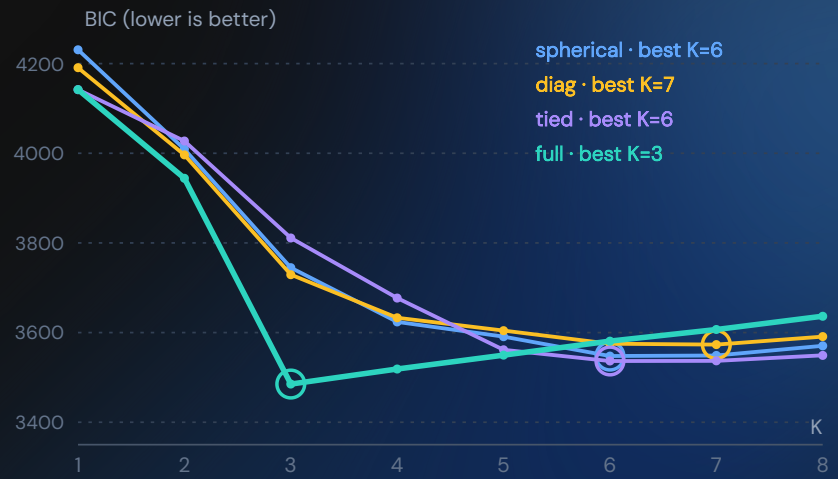


$$\mathrm{BIC} = -2\log\hat{L} + p\log n, \qquad \mathrm{AIC} = -2\log\hat{L} + 2p$$

K = 8 · p = 47 · BIC 3636 · AIC 3438

K = 8: −2 log L still falling; BIC clearly worse.

Choose K and Covariance Type Together



full, K = 3 wins: BIC 3485.

Constrained shapes want **more components**: K = 6–7, stacking small round or axis-aligned blobs to imitate tilted ellipses.

Grid over both, keep the lowest BIC. Number of components ≠ number of "real" clusters when shapes are wrong.

Gaussian Mixtures in scikit-learn

```python
from sklearn.mixture import GaussianMixture

gmm = GaussianMixture(n_components=3, covariance_type="full",
                      n_init=1, random_state=0)  # init: k-means
gmm.fit(X)                        # X: (500, 2), no labels
gmm.converged_, gmm.n_iter_       # (True, 6)
gmm.weights_.round(2)             # [0.43 0.37 0.21]
gmm.means_.round(2)               # [[0.06 0.06] [4.34 1.27] [2.56 -2.54]]

P = gmm.predict_proba(X)          # (500, 3) responsibilities
P[[0, 1, 423]].round(2)           # [[0.07 0.93 0.  ]
                                  #  [0.   0.   1.  ]
                                  #  [0.54 0.46 0.  ]]
gmm.predict(X[[0, 1, 423]])       # [1 2 0]   argmax of each row
gmm.bic(X), gmm.aic(X)            # (3485.2, 3413.5)
gmm.score_samples(X[:2])          # log p(x): [-3.1 -3.71]
```

`predict_proba` : the γ table. Row 423 is a boundary point: **0.54 / 0.46**.

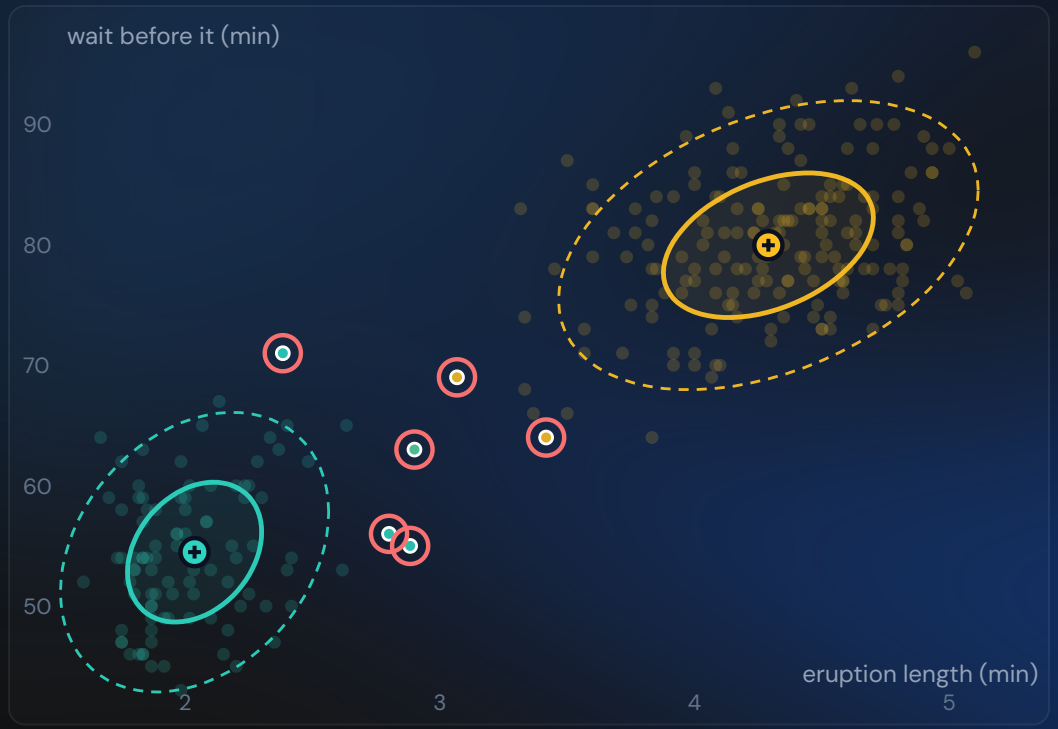
`score_samples` : log-density of each point, for anomalies. `sample(n)` : new data.

Set `n_init` > 1. Scale features first: a Gaussian's shape is in the units you give it, though full Σ absorbs per-feature scale better than k-means.

Part 5 · Real Data, and What a Density Buys You

A geyser, a garden, and two moons

Old Faithful: Two Kinds of Eruption

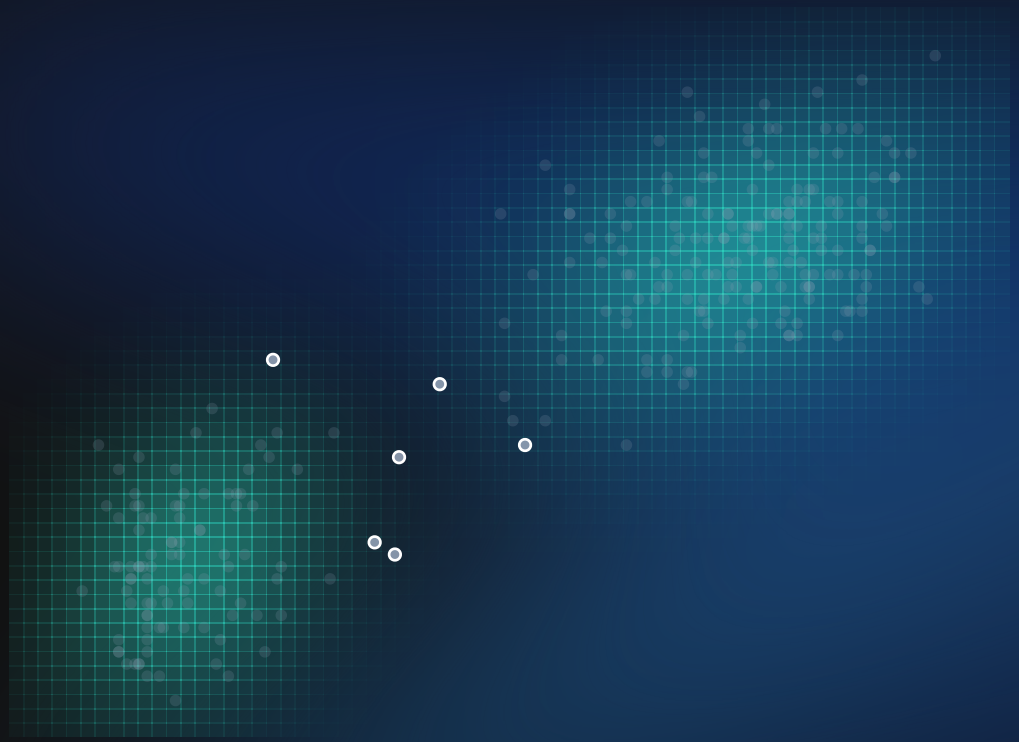


$K = 2 \cdot \pi = 0.36 / 0.64$

short eruptions ≈ 2.0 min after a 54 min wait; long ≈ 4.3 min after 80 min

6 unusual eruptions: log p(x) < −7.3, the lowest 2%. Mostly mid-length eruptions, between the two types.

The Density Landscape and Its Anomalies



Two ridges, one per eruption type. Each eruption is lifted to its own density.

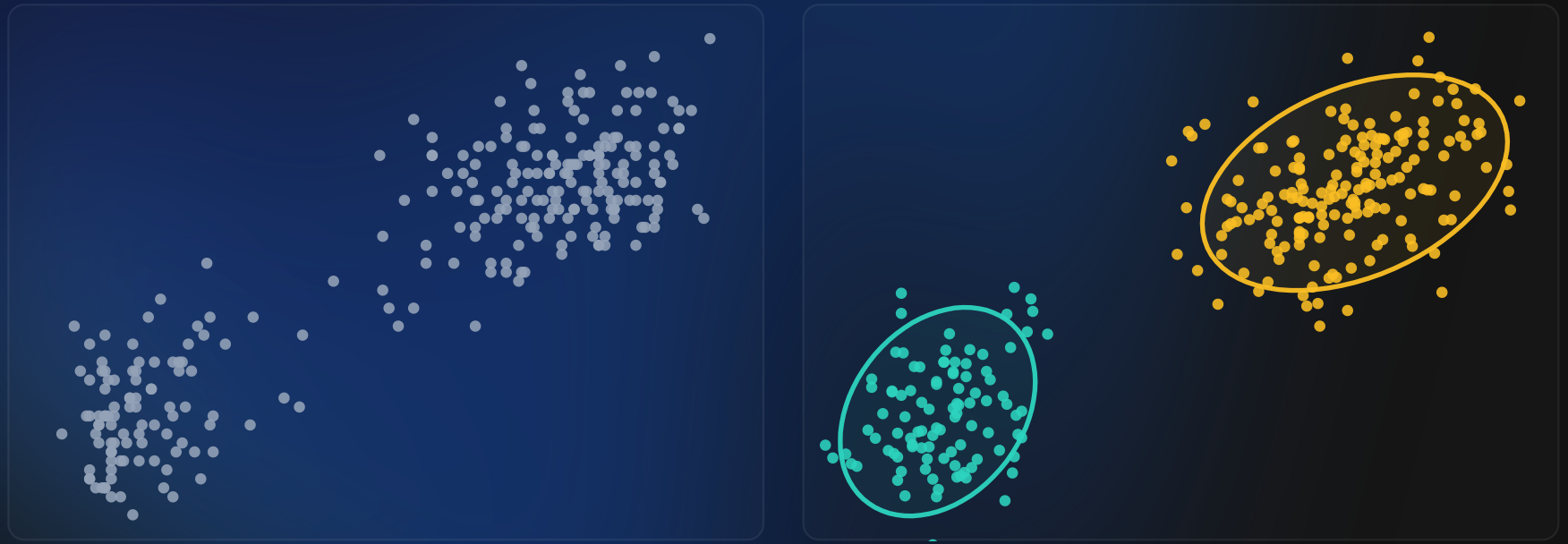
Red plane: the density threshold. The 6 red points fall below it.

```
logp = gmm.score_samples(X)
cut = np.percentile(logp, 2)
anomalies = X[logp < cut]
```

A GMM Can Generate New Data

Real: 272 eruptions

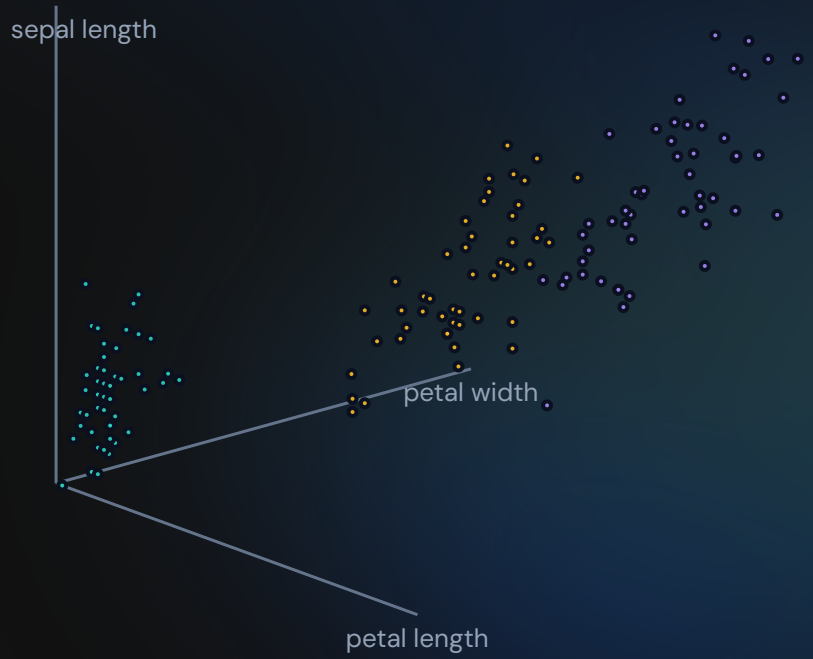
Sampled from the GMM: 272



```
X_new, comp = gmm.sample(272)
```

Run the generative story forward: spin the wheel, draw from that Gaussian. Synthetic data with the same structure, the simplest **generative model**.

Iris in 3D: Ellipsoids vs. Spheres

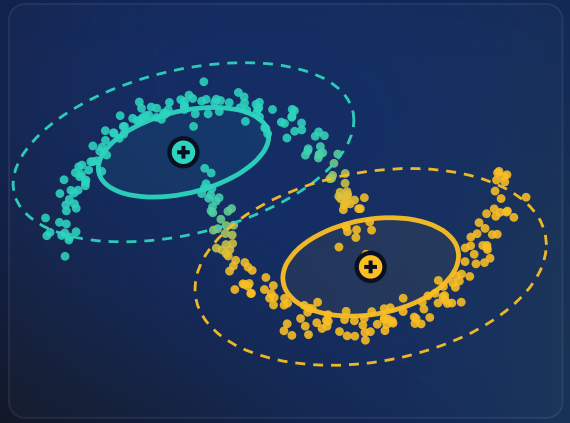


150 flowers · petal length, petal width, sepal length (cm). Colors: **GMM, K = 3, full covariance**

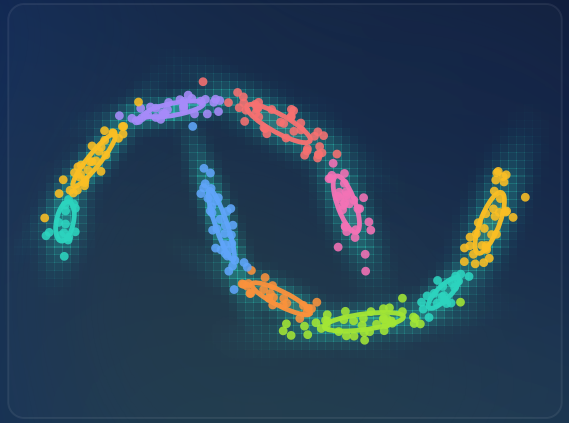
k-means · ARI 0.73. Cuts the two overlapping species with a flat plane.

GMM · ARI 0.94. Elongated, tilted ellipsoids follow each species. (All 4 features: 0.90 vs. 0.73.)

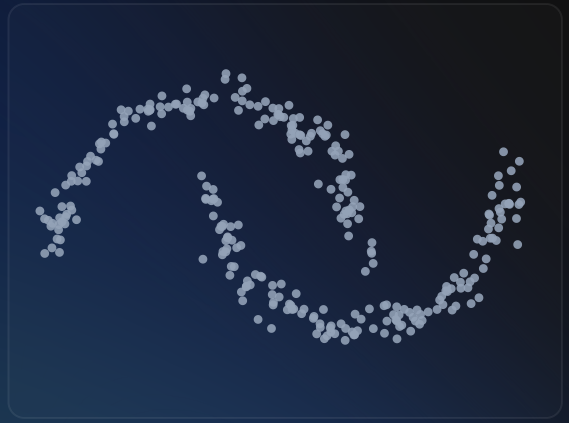
Where GMM Fails: Shapes That Aren’t Ellipses



K = 2 · ARI 0.46



K = 10 · density fits · as clusters, ARI 0.21



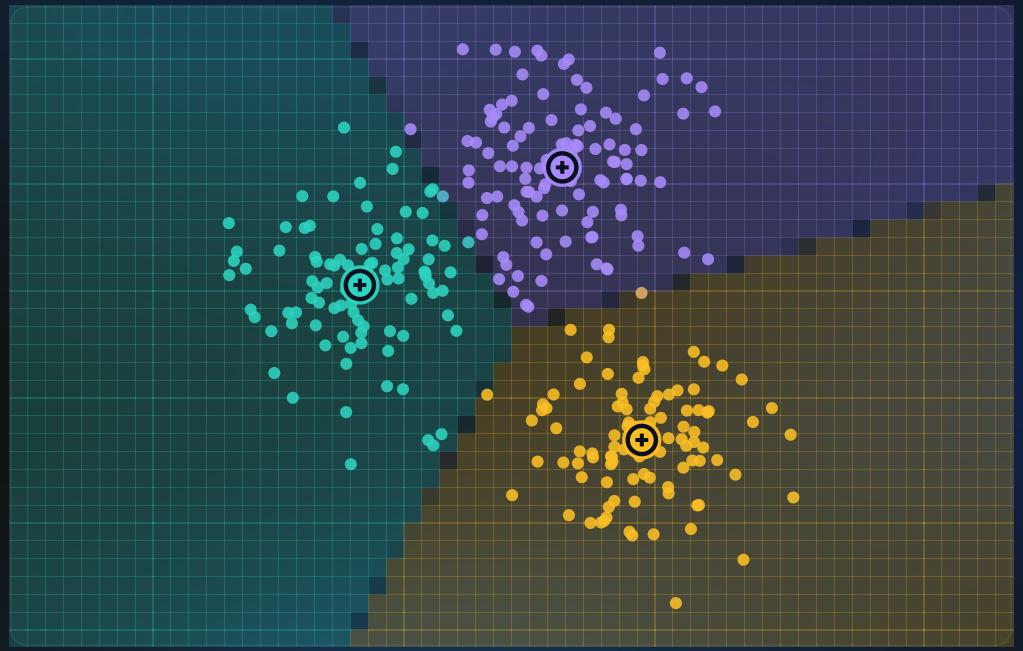
Samples from K = 10 look like moons

A Gaussian is convex. Curved clusters → use DBSCAN/HDBSCAN for labels; a many-component GMM is still a fine **density**.

Part 6 · The Big Picture

k-means was a Gaussian mixture all along

k-Means Is a GMM with the Knobs Turned Down



σ² = **0.050**

points with max γ < 0.99: **5**

mean entropy: **0.006**

$$\pi_k = \tfrac{1}{K},\ \Sigma_k = \sigma^2 I: \quad \gamma_{ik} = \frac{e^{-\|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2/2\sigma^2}}{\sum_j e^{-\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2/2\sigma^2}} \xrightarrow{\sigma^2 \to 0} 1\left[k = \arg\min_j \|\mathbf{x}_i - \boldsymbol{\mu}_j\|\right]$$

E → nearest center. M → plain means. **EM = Lloyd's algorithm.**

Choosing a Clustering Method

	CLUSTER SHAPE	NEEDS	MEMBERSHIP	SCALES TO	REACH FOR IT WHEN
k-means	round, similar size	K	hard	millions	you need K prototypes or a fast partition
Hierarchical	depends on linkage	linkage, cut	hard	~10^4	the tree is the answer, or a custom distance
DBSCAN / HDBSCAN	any shape	ε, minPts / min size	hard + noise	~10^6	curved shapes, outliers, varying density
GMM	ellipses, any size/tilt	K, covariance type	**probabilities**	millions (low d)	overlapping clusters, uncertainty, a density for scoring or sampling

GMM's extra costs: more parameters (watch d), local optima (use `n_init`), and K chosen by BIC rather than read from the data.

Summary

A cluster is a Gaussian

Mean = center; covariance = an ellipse. Distance is Mahalanobis, in the cluster's own shape.

Mixture = generative story

Pick k with probability π_k, draw from its Gaussian. Responsibilities γ = Bayes' rule, read backwards.

EM

E: soft labels. M: weighted means, covariances, weights. Likelihood never drops; local optima → n_init.

Model choice

Covariance type trades flexibility for parameters. Pick K (and type) by lowest BIC.

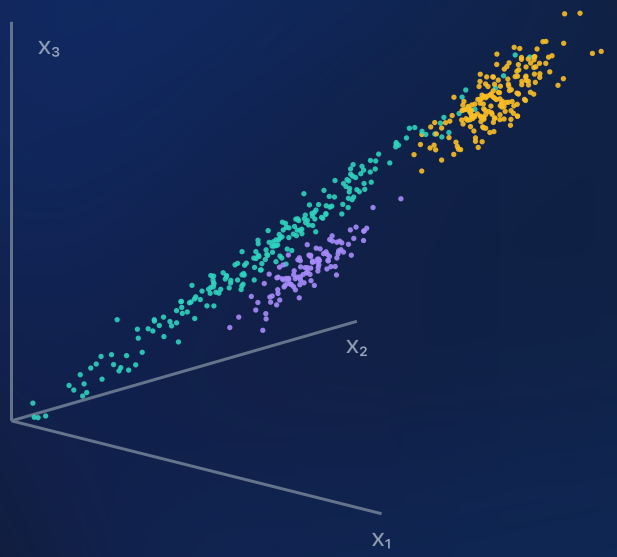
A density, not just labels

score_samples for anomalies, sample for new data. Components ≠ clusters for curved shapes.

k-means is a special case

Equal weights, shared $\sigma^2 I$, $\sigma^2 \to 0$: EM becomes Lloyd's algorithm.

Next: Dimensionality Reduction



The problem: full-covariance GMMs need ~$d^2/2$ numbers per component, and distances lose meaning in high dimension.

Next deck: these 3D points really live on a tilted plane. PCA finds the few directions that matter, and its axes are the **eigenvectors of a covariance matrix**, today's ellipse axes. Then t-SNE and UMAP for visualizing structure.